

Performance Testing

Date	07 July 2026
Team ID	
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	Prediction Engine API
Test Environment	Local
Test Date	2026-07-07 T00:00:00

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Prediction API Request	1	5	Successful prediction
2	Batch CSV Upload Processing	10	30	Successful batch processing

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.4s	Pass	
2	Response Time (Max)	< 5 seconds	1.2s	Pass	
3	Throughput (Req/sec)	N/A	50 req/s	Pass	
4	Error Rate	< 1%	0%	Pass	
5	CPU Utilization	< 80%	35%	Pass	
6	Memory Utilization	< 80%	42%	Pass	

Step 4: Observations & Analysis

- **Key Findings:** The application handles single and batch predictions efficiently within the target response time limits.
- **Bottlenecks Identified:** Minimal latency observed during large batch CSV uploads.
- **Optimization Steps Taken:** Implemented asynchronous processing for batch CSV uploads to improve throughput.